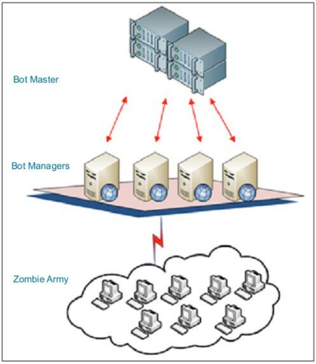

Figure 1: Botnet

table below explains various types of botnets and the purpose behind injecting them into a network. The overall purpose behind such an attack is, ultimately, to disrupt computer systems or to steal data. Since a whole army of computer zombies are in action, unfortunately, the crackers can easily and quickly succeed in their evil mission; this is because planting a botnet attack is always a low-risk, high-profit job.

Botnet type	Purpose
DoSBot	DoS and Distributed DoS attack using Layer 3 to 7 protocols
SpamBot	Email spamming by collecting address books
BrowseBot	Gather user's browsing trends and feed into advertisement network
AdSenseBot	Same as *BrowseBot* but targeted at Google AdSense
ChatBot	Collect chat transcripts to find user's chatting trends
idBot	Collect user ID and password information
CCBot	Collect credit-card information from e-commerce portal screens
PollBot	Manipulate online polls meant for products and services
BruteForce-Bot	Attack websites with TCP and application layer attacks
NetBot	Attack networks using Layer 2 and 3 protocols

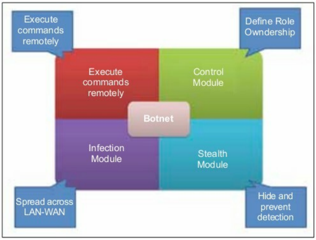

Figure 2: Modules of a botnet

How botnets are injected

In the early days of the Internet, a botnet code piece was developed to programmatically traverse through multiple websites, and to further gather and collate the contents to create meaningful data. While this method forms the heart of today's search engines, it was tweaked at some point in the past by crackers to serve their purposes. Before discussing how botnets are injected, let's understand why it is done. To make a website famous in a search engine, it is imperative to get lots of Web requests. This is especially true for websites that run advertisements and earn money for every click on a published advertisement. It is now possible to spread botnets across the networks, to access the page and programmatically click one or more advertisements on it. If such a campaign is carefully orchestrated, it is tough to figure out which click is legitimately initiated by a human being, and which one originates from botnet code. The website hosting firm, usually a cracker in such a case, can end up earning lots of money. In another type of attack called phishbots, an email campaign can be started to achieve similar results. This tells us that the effects of botnets go much beyond mere reputation or data loss.

Injecting a botnet is usually a very well-thought-out and strategic approach taken by the cracker. The process usually starts by infecting one or more systems, which are then responsible for replicating the malicious code in other machines, and eventually they cross the boundaries of the network to spread the infection to a wider global arena. In order to infect one system, the attacker needs to rely on multiple methods of intrusion. A very commonly used option is to lure a browser to a website with malicious JavaScript code, or a page written in a low-level scripting language such as Python. This script is merely a bootstrap, which executes and creates a stealth resource space on the machine. The script then connects to one or more Web pages of the same website, which contain the real payload of a botnet. The payload files are then downloaded and kept hidden under a stealth space. This payload contains all the modules explained above, which take control of the machine, and the machine is said to be infected at this point. Enhanced botnets do not require the